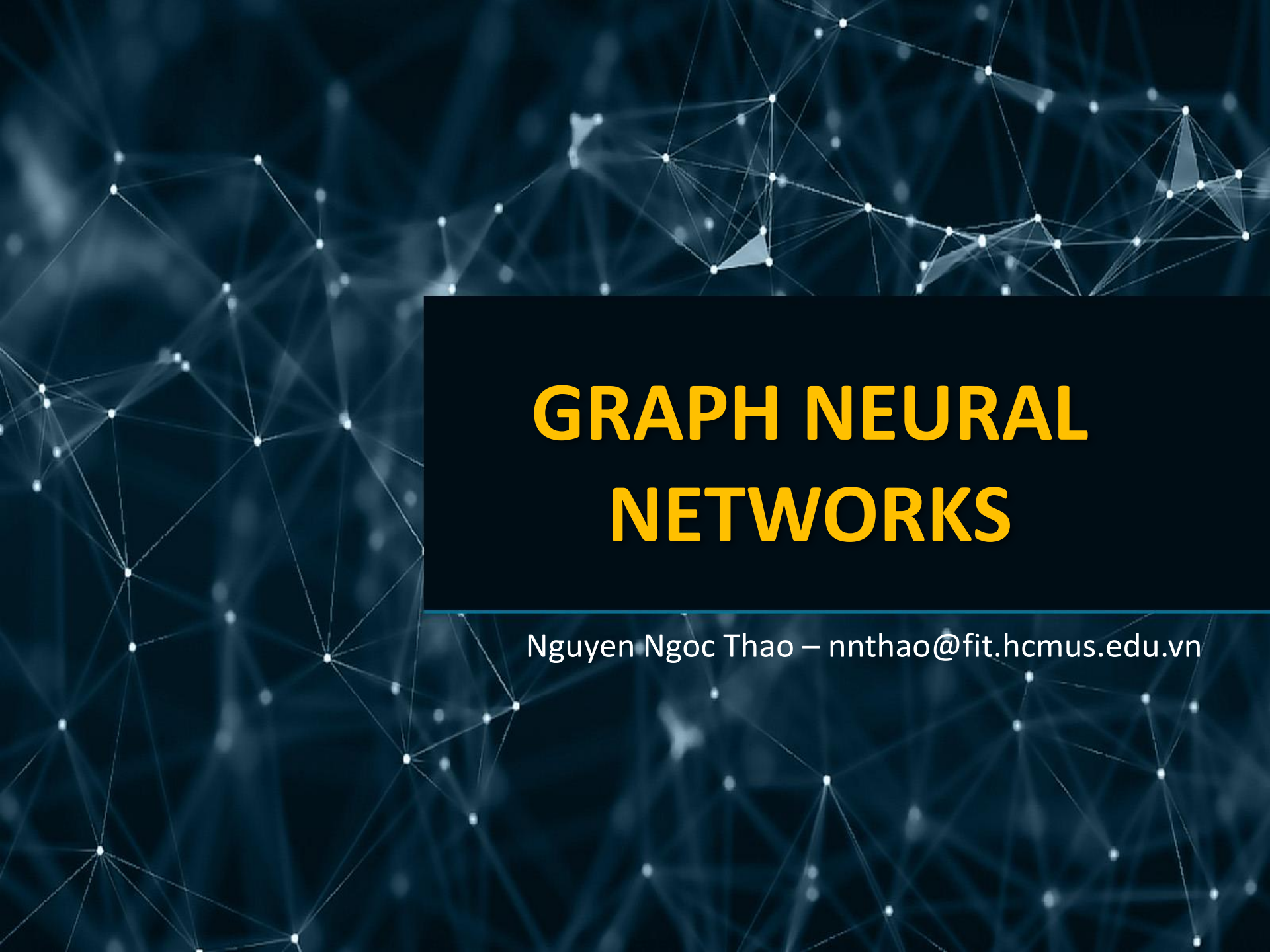




GRAPH NEURAL NETWORKS

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Outline

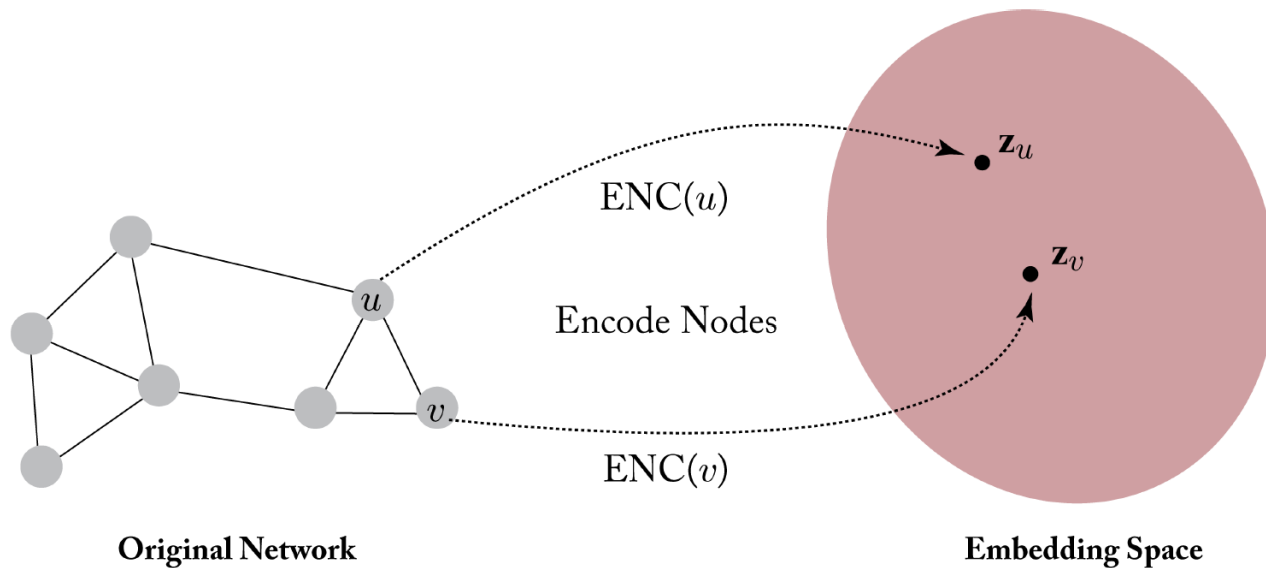
- Graph representation learning



Graph representation learning

Graph embeddings: A definition

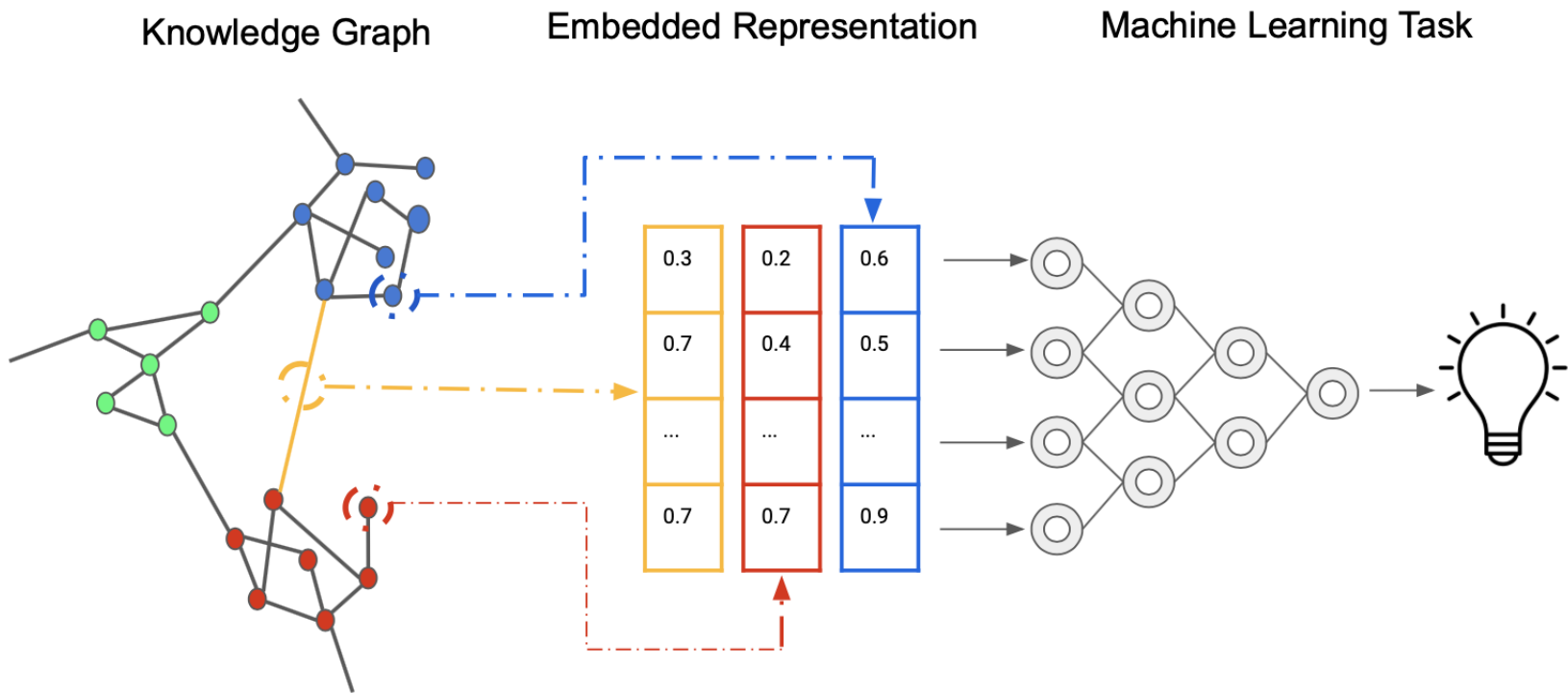
- **Graph embeddings** transform the **entities and relations** in knowledge graphs into **low-dimensional numerical vectors**.



High-dimensional nodes are mapped to a low-dimensional space via a set of transformations learned through an encoder (ENC).

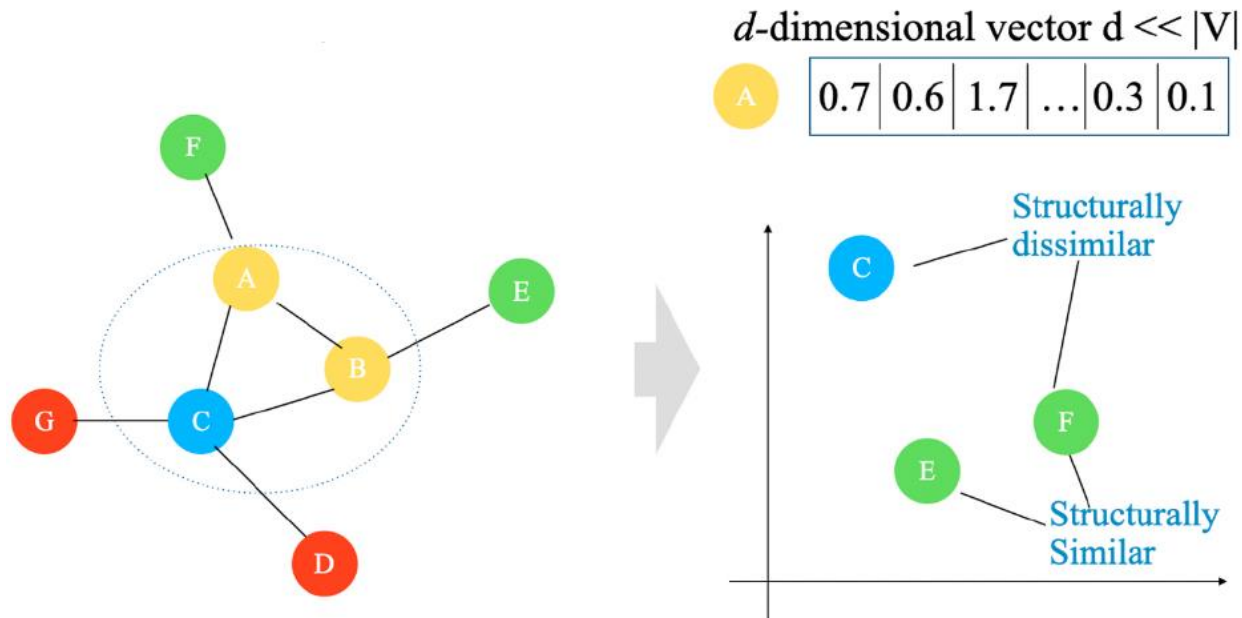
Graph embeddings: Applications

- They are used as inputs for several machine learning tasks.
 - E.g., node classification, link prediction, recommendations, etc.



Graph embeddings: Quality

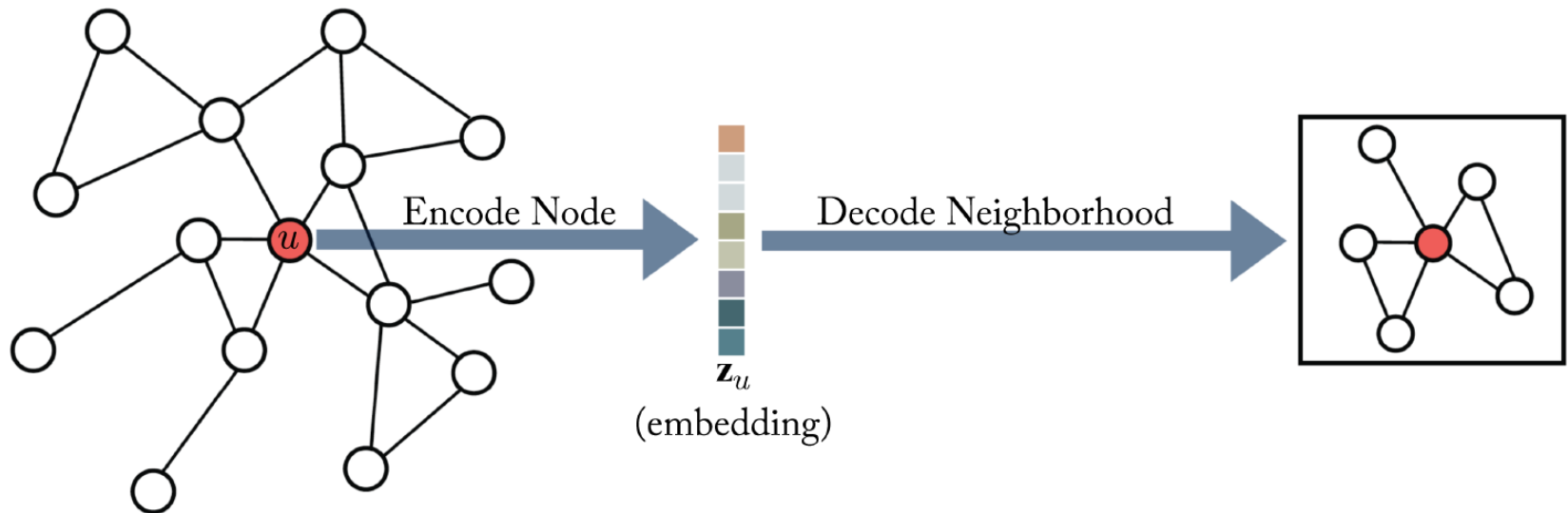
- A good graph embedding should preserve structural and relational information, while minimize redundancy and noise in the original data.



Embedding dimension d is usually much smaller than the number of nodes $|V|$ in the graph.

An encoder-decoder perspective

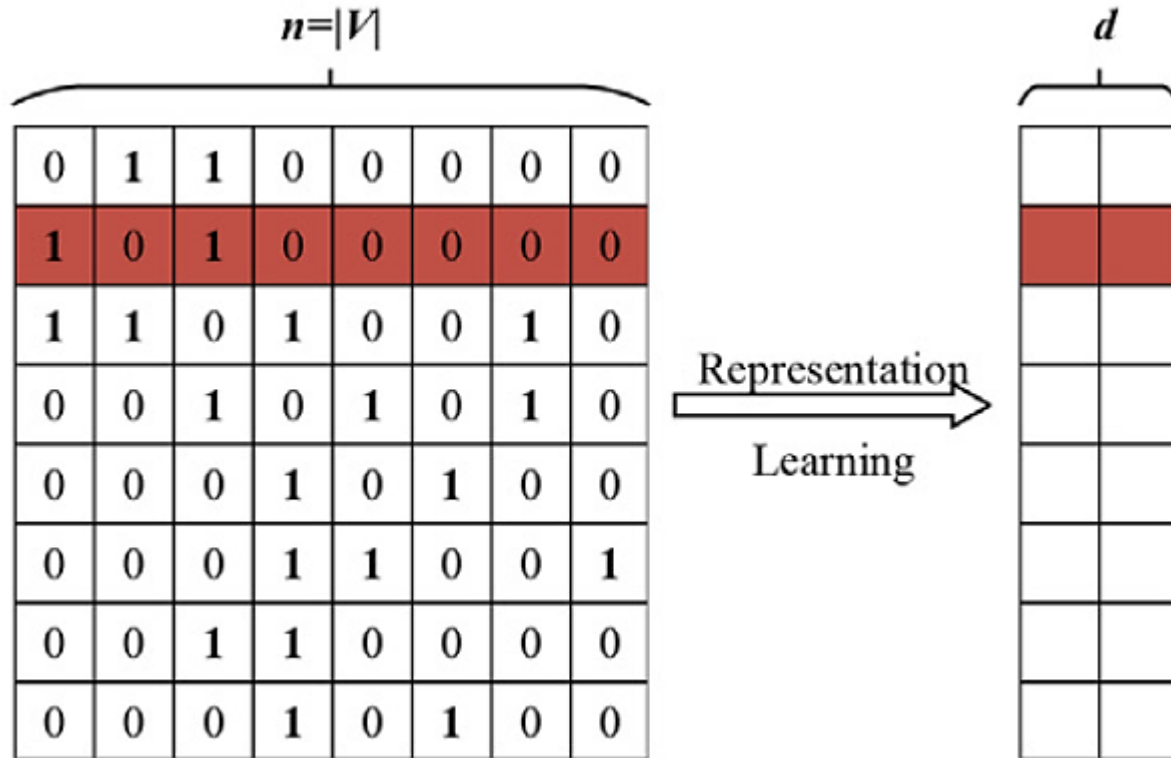
- An **encoder-decoder framework** for graph representation learning is built using **neighborhood reconstruction**.



The encoder maps nodes into a low-dimensional vector, and the decoder uses this vector embedding to reconstruct original information about nodes' neighbors. The system is trained using an appropriate reconstruction loss (such as mean-squared error, or cross-entropy).

Factorization-based models

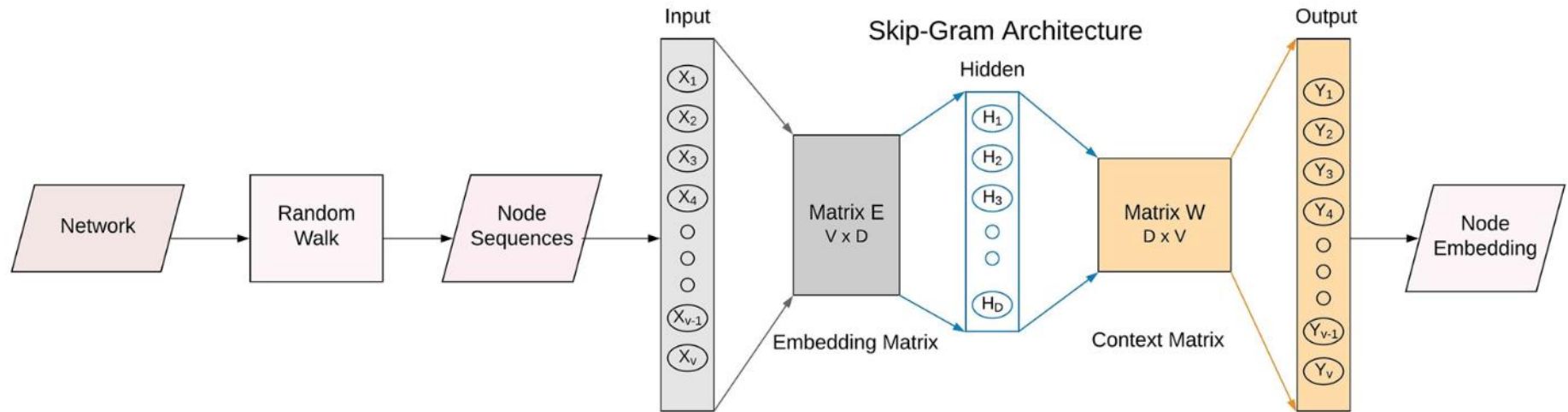
- Factorize simple graph representations (e.g., adjacency matrix w/wo weights) to generate node embeddings.



Random-walk based models

- A **random walk** represents a sequence of nodes where the **next node is selected** based on a **probabilistic distribution**.
- **Goal:** Extract rich structural patterns from the graph's local structure.
- **Objective:** Maximize the probability that nodes co-occurring on truncated walks are close in the embedding space.
- It may backtrack or visit the same node multiple times.

Random-walk based models



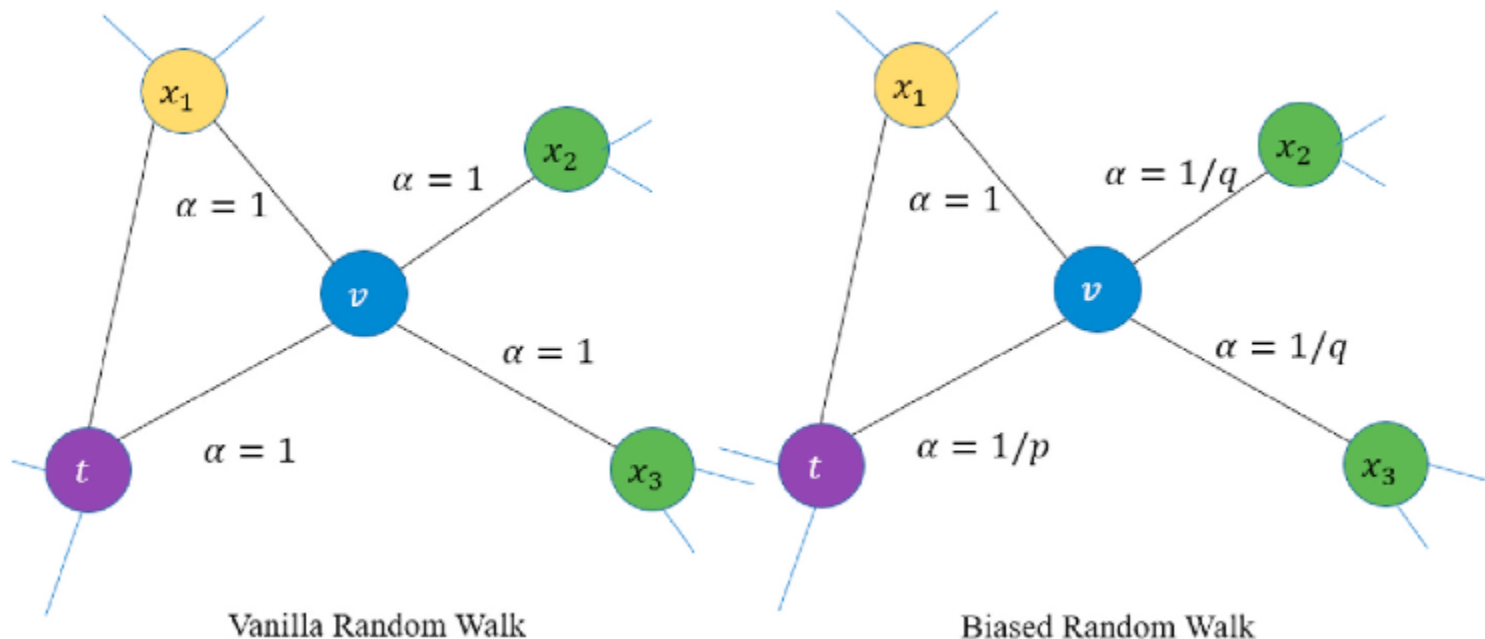
First, perform a truncated random walk to transform graph input to node sequences.

Next, a skip-gram model, which is a shallow neural network, maximizes the conditional probability of association between each input and its surrounding context.

Finally, node embeddings are learned by using each node to predict its context.

Random-walk based models

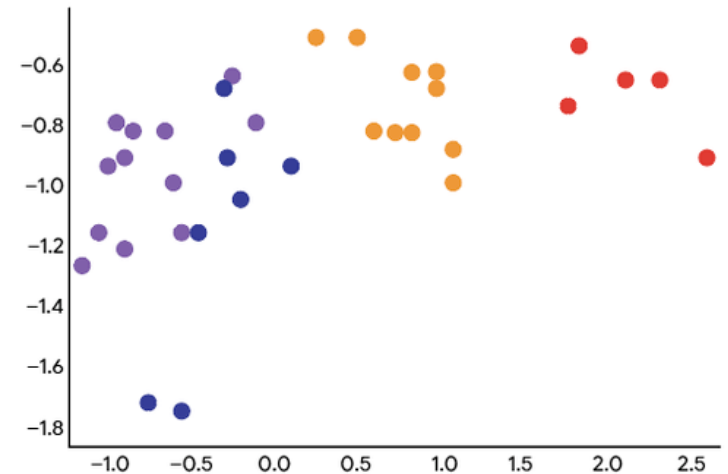
- [DeepWalk](#) was one of the first models based on random walk.
- [node2vec](#) further improved it by replacing the unbiased walk with a biased walk.
 - It uses two parameters that control the probability of returning to the previous node and performs either a BFS or DFS.



Random-walk based models



Input: Graph



Output: Representation

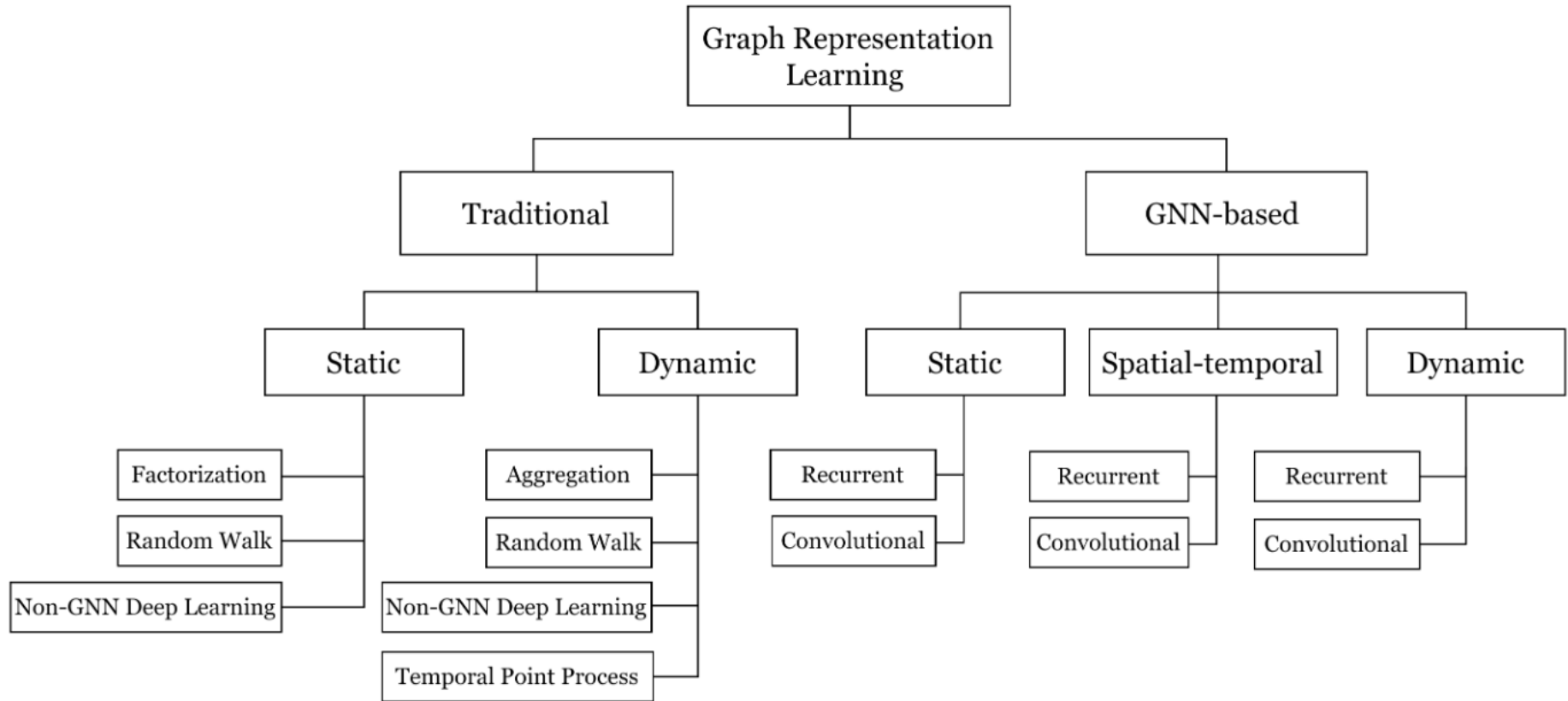
DeepWalk learns a representation that encodes graph information.

Perozzi, Bryan, Rami Al-Rfou, and Steven Skiena. "DeepWalk: Online learning of social representations." In Proceedings of the 20th ACM SIGKDD international conference on Knowledge discovery and data mining, pp. 701-710. 2014.

Deep Learning based Models

- The limitations of conventional models
 - The embedding is learned from a model and is then fed to a downstream model for a task.
 - Shallow neural embeddings often lead to suboptimal solutions because they can not effectively model the nonlinearity that is inherent in network data.
- Deep learning enables end-to-end solutions by inherently learning the embeddings in the architecture.
- Graph Neural Networks (GNNs) use the Message Passing technique to effectively encode information about the graph.

Graph embeddings: Taxonomy



A taxonomy for graph representation learning

Khoshraftar, Shima, and Aijun An. "A survey on graph representation learning methods." ACM Transactions on Intelligent Systems and Technology 15, no. 1 (2024): 1-55.

...the end.

